

Mahesh Paka

LinkedIn: [https:// www.linkedin.com/in/mahesh-paka-jan115](https://www.linkedin.com/in/mahesh-paka-jan115)

GitHub: <https://github.com/Maheshgitt>

Email: maheshpaka115@gmail.com

Mobile: +91-8464985128

SKILLS

- **Languages:** C++, Java, Python, Kotlin, Verilog
- **Frameworks:** Simulink, Anaconda Prompt, DSCH2, Control System Toolbox, Arduino Libraries
- **Tools/Platforms:** Arduino IDE, Android Studio, Spyder, MATLAB, Proteus, Vivado, Keil, FastAPI, Firebase, Render.com
- **Soft Skills:** Team Player, Project Management, Persistent, Problem-Solving, Adaptability, Self-Motivated

TRAININGS

Internshala

Android App Development

Jul' 2025

- Completed online training in Android App Development with hands-on experience in UI design, Firebase, API integration.
- Built and tested Android applications using Kotlin and Android Studio.
- Gained experience in Jetpack Compose, Retrofit, and the mobile development lifecycle.

Tech Stack: Kotlin, Android Studio, Jetpack Compose, Firebase, Retrofit, APIs

Embedded Systems

May' 2025

- Completed online training in Embedded Systems with microcontroller programming and circuit interfacing.
- Designed mini-projects with data acquisition, UART/I2C/SPI protocols, and IoT integration.
- Gained experience in embedded C, debugging, and simulation using Proteus and Keil.

Tech Stack: Embedded C, Arduino IDE, ESP32, Keil µVision, Proteus, Sensors & Actuators

PROJECTS

AI Tech News – Android Application

May' 2026

- Built a full-stack AI news app with Python FastAPI backend using Groq LLMs for intelligent article summarization.
- Integrated NewsData.io for live feeds and Firebase Cloud Messaging for real-time push notifications.
- Deployed on Render.com with a Kotlin/Jetpack Compose frontend following MVVM architecture.

Tech: Kotlin, Jetpack Compose, Python, FastAPI, Groq LLMs, NewsData.io, Firebase FCM, Render.com

Aegis Node – ESP32 Industrial Safety Wearable

Apr' 2026

- Designed an ESP32 wearable integrating MAX30102, MPU-6050, and MQ-2 sensors with OLED display and buzzer alerts.
- Built a live WebSocket dashboard with Chart.js waveforms for real-time SpO2, heart rate, and gas monitoring.
- Implemented fall detection via accelerometer magnitude thresholding with tiered safety alert system.

Tech: ESP32, C++, MAX30102, MPU-6050, MQ-2, WebSocket, ESPAsyncWebServer, Chart.js

BER Checker – FPGA Implementation

Apr' 2026

- Implemented a 4-bit Bit Error Rate checker in Verilog on Nexys 4 Artix-7 FPGA with 7-segment display output.
- Designed modular RTL with ber_checker and seg_decoder modules verified using a dedicated testbench.
- Synthesized and validated the design using Xilinx Vivado with timing analysis and simulation waveforms.

Tech: Verilog, Xilinx Vivado, Nexys 4 Artix-7 FPGA, RTL Design

CERTIFICATES

- NPTEL/Swayam Data Base Management System Sep' 2025
- NSDC/Internshala- Android app Development Jul' 2025
- NSDC/Internshala-Embedded systems May 2025
- Introduction of Python in scalar Jan' 2025

EDUCATION

- **Lovely Professional University** Punjab, India
Bachelor of Technology – Electronics Communication and Engineering; CGPA: 7.55 Since August 2023
- **Sree Sai Junior College** Hanamkonda, Telangana
Intermediate; Percentage: 97.8% June 2021 - May 2023
- **MJPTBCWREIS School** Hanamkonda, Telangana
Matriculation; Percentage: 100% June 2020 - April 2021